E–Learning and Digital Media
Volume 11, Number 5, 2014
www.wwwords.co.uk/ELEA

e-Learning and Digital Media

ChipScope: actually, that funny way of looking at it works pretty well

KEVIN M. LEANDER
Department of Teaching and Learning,
Peabody College, Vanderbilt University, Nashville, USA

ABSTRACT In this article, the author describes and reflects on a kind of vision that is unique to Bertram C. ('Chip') Bruce. This 'funny way of looking at it' includes a way of seeing the world as it is made in myriad connections, a way of seeing the world imaginatively and metaphorically, and a way of seeing the world with optimism and empathy.

I have to say that as I started to think about what I wanted to share today I had a lot of questions in my mind:

- How do I say something and make it truly about Chip, and not about myself? Or is this a dialogue of lives and influence?
- How do I write this without making it sound like Chip's funeral?
- Is there any way to make this not an idealization, but rather a situated evaluation of Chip?
- How well do I really know Chip?
- What is a Festschrift, and is there any way we can have one and make it feel less German?

And most importantly:

- What would Chip do?

My answer to the last question was actually pretty helpful because I knew what Chip would do. While I would approach this writing task by deciding I knew nothing, or nothing very certain at all, and it would probably take me years of research to say anything useful at all, Chip would say: 'Actually, I think I know quite a lot about that already. I wonder what I do know?' And, without further ado he would proceed to write, and develop page after page without having gone anywhere or gathered anything up, just quite yet. And he would prove to himself again that actually, he did know quite a lot about that already.

So, I think what I'd like to do in this brief time is to share one artifact and a few stories, in order to give some small sense of what it is like to be one of Chip's doctoral students, and what it's like to see, from the periphery, to see how he sees.

So, first, the artifact. I'm lucky enough to still have in my possession the page proofs of a paper published in 1997 by Chip and several others on ChickScope, a project that many of you might be familiar with and that involved, as far as I could tell then, letting kids and teachers control an MRI scanner remotely, through the web, to look at developing chick embryos in real time. These page proofs tell the story of a scientist who is imagining things outside the norm – who is courageous and not shaken by struggles, and who inserts himself into that brave territory where school meets the university meets innovation.

I'll focus on just a single marginal comment from the editor on pages 80-81, but first allow me to read a bit of the text from that section. It says: 'Once a week, when the entire class used the computer lab, the students had the opportunity to browse through the ChickScope Web site to read Today's News, search the MRI database for acquired images, review Carl's Roost, or share a

http://dx.doi.org/10.2304/elea.2014.11.5.471

chicken joke'. Further on it says, 'The teacher and the student teacher would regularly share the scientists' responses or answers with the entire class for discussion purposes. Here are some examples of such interactions which were useful to all the students in this classroom (as well as in other participating classrooms):

> Q: Why is the MRI image in black and white?
>
> A: Good question! Actually, the MRI image has no 'colors' at all – it is a radio signal.
>
> But in order to make it visible, the computer artificially colors it. Since a single image conveys only one value at each point, we choose to assign that value to 'brightness'.
>
> Q: Why can't we see anything except for weird blobs and fuzzy stuff?
>
> A: Good question! If you squint your eyes a bit, though, you will notice that it looks like a chick embryo. Squint harder.
>
> Q: Why is the computer lab in the school closed on Mondays, Wednesdays, and Fridays, and yet the MRI is open every day?
>
> A: Good question! We are working on asking the schools to see if they would be willing to allow you to use their technology, but this is tough business.
>
> Q: Why does that one guy from the university have such corny chicken jokes?
>
> A: No one knows the answer to this question, not even scientists.'

So, the only, but key marginal comment I have on the page proofs in this section, from the editor is: 'Please cut back some of these questions. Too many'.

So, ChickScope is a way of course about talking about ChipScope – about the ways of an unbounded optimist who will always say it's never just about the technology, or who always sees the technology in the everyday connections and conundrums of people and policies and power relations and social practices that sometimes extend over decades, yet sometimes can change, seemingly, in minutes.

ChipScope is also a way of looking up close at the world, but seeing things differently. One particularly annoying but endearing trait of Chip and his scope on things is that it's almost impossible to come up with a joke about how things could be done that he can't appropriate into a possibility.

If you don't believe this, you should try it in a thought experiment, or perhaps in a real experiment. For instance, start out by saying something crazy like, '*Dewey was mainly a technologist'*. No, actually, not that. Try a game of crazy metaphors in your mind. Picture yourself in a conversation where you are talking about the federal government with Chip and others. And everyone is talking about how it's a slow moving machine – the cogs of the wheels turn, it reproduces the same things, its powerful and predictable, all that. Now, be capricious and introduce into that dialogue a non sequitur: the government is actually just like a potato, or a pickle, or still better, a sea urchin.

And Chip will say, 'You know, actually, that funny way of looking at it works pretty well if you think about it. From what I understand about sea urchins, they create exactly the same chemistry inside their bodies as found outside in the sea. That's interesting for understanding governments. Also, sea urchins are one of the only creatures that can only be eaten by one predator – the sea lion – and, interestingly, sea lions appear to choose the smallest governments – I mean, sea urchins – for their diet. So it's all connected. In fact, that works great. I think I'll write an article'.

It seems impossible, really, to see things too strangely – to make a metaphor too awkward to open at least one beer bottle of ideas around ChipScope.

So, Chip's approach to sea urchins set up a context in which I was encouraged to think about critical discourse analysis, bodies, and emerging theories of social space. Why would someone read critical and urban geography as a way of thinking about discourse in school classrooms? Chip got it. More sea urchins.

Now, for another non sequitur …

I think it was 1993 or early 1994. I was a master's student in writing studies, working on a master's degree in the English department. I was enrolled in a course with Chip in education that was very formative for my own intellectual development. The course was completely unusual for curriculum instruction because it was invested in theory and because classrooms didn't figure as the prominent or unique site for thinking about learning and development. In this course, Discourses of Science, we read Latour and Lynch and Woolgar and Feyerabend and Foucault and a whole host of others. Most of the thinkers that Chip introduced me to in that course are people I still want to think with.

Four years later after taking the course, I was presenting my first paper at the Society for the Social Studies of Science annual meeting. It was a paper on media reporting about Biosphere II.

Some 18 years after taking the course, I'm a co-PI on an NSF funded project that studies how scientists and professionals use spatial analysis and modeling, supported by new technologies, in their everyday practices.

So, that's a digression about effects and intellectual influence. What I really wanted to get to happened right after class one day …

Chip told a few of us to stop by his office. For some reason or other, the College of Education had decided to build phone booths for faculty offices. As doctoral students, we imagined that the professors changed their clothes in them to become superhero academics. In any case, Chip invited 3 or 4 of us to come into his office and see something on this computer screen. He said, 'Look at this – this is pretty cool'. We had no idea what he was showing us. What I remember seeing was that Chip clicked on the keyboard a few times and this colorful circle in the corner of his screen appeared to turn round and round, like a globe. And then the circle froze and nothing happened. Chip paused, sighed, and said, 'Okay, you'll have to come back sometime. It's pretty cool'.

So, that was the World Wide Web. That day, it wasn't cool. But some day it might be. So, one thing I've learned about Chip's scopic vision over time is that you should trust those little hints that something might be worth checking out. If he drops hints, like he did back in 1994 with his prescient article entitled 'Dewey was Mostly Interested in Global Warming', the rest of us should really take notice. Seriously.

But I want to close with a thought that is not about what Chip sees or what it means to look closely or to look broadly, or to have inter-disciplinary vision that extends beyond fancy tools and tasks. Rather, I'd like to talk for a minute about Chip's hopeful and intentional vision – what it means to see what one *wants to* see, not as a defect, but as part of a *belief that seeing things is making things*.

So, how do you see in order to make? And in particular, how do you see people and their ways in order to remake the world? Here's a scenario that comes to mind:

> I'm a third year doc. student, waiting outside of a seminar room that I need to use for a committee meeting of some kind – I'm meeting about some proposal with other faculty. I'm nervous. I've reserved the room. I stand outside the door, waiting for a meeting running late, and one faculty member inside the room, in that meeting going on and on makes a smart remark about what on earth I'm doing there, standing in the hall. It's one of those jokes that is funny and not funny, a joke that makes one feel small, even smaller than a doctoral student.

Afterward, the silly little put down joke makes me angry and I tell Chip about it. So, how is seeing things making things, making the world?

I really want him to say that the room was filled with mean bad people. I want Chip to identify the bad people with me as people who need to be fixed. I'm pretty sure I know how to fix them. But instead, Chip talks about how jokes like that don't feel like jokes, he talks about how rooms get scheduled and how time works to make institutions function, he talks about how people respond in moments without much awareness. He's empathetic, thoughtful, reasonable, but unwilling to blame. Dammit, I think. He did it again.

So, I think that's a good example of choosing to see, and choosing to see with courage. It is, after all, what we have: these choices that we make about the world, these ways of not just seeing, but re-envisioning.

Seeing again, and making differently, one more time. Actually, that funny way of looking at it works pretty well.

KEVIN LEANDER is Associate Professor of Language, Literacy and Culture at Peabody College of Vanderbilt University, USA. His research interests include the new literacy practices of youth, spatial approaches to understanding youth identity and learning, research on new media, and media and migration. Leander was a project leader in 'Wired Up' at the University of Utrecht (with Mariette de Haan and Sandra Ponzanesi) and is also a leader in the 'Space, Learning, and Mobility' (SLaM) group at Vanderbilt, engaged in NSF-funded research on spatial thinking (with Rogers Hall). Leander is most recently engaged in thinking about and designing new learning environments, including hybrid environments that traverse online and physical spaces. Leander has published widely in journals such as *Review of Research in Education, Ethos, Reading Research Quarterly, Journal of Literacy Research,* and *Cognition and Instruction.* He has also authored and co-authored handbook chapters on youth and new media, multimodality, and mobile technologies. *Correspondence*: kevin.leander@vanderbilt.edu